DRIP PROTOCOL

Milestone & Evaluation Document

Fhenix Privacy-by-Design Buildathon — Wave 1 through Wave 5

Milestone Philosophy

Drip Protocol milestones are structured to deliver a working, testable protocol at the end of every wave — not just a spec or a design doc. Each wave targets a shippable layer of the stack. Evaluation criteria are verifiable on-chain: a testnet address, a transaction hash, a test suite output, or a live demo.

The FHE privacy layer is integrated from Wave 1 — not retrofitted at the end. This reflects the core thesis: privacy-by-design means every contract is built with encrypted data types from day one.

Programme Overview

Wave	Phase	Dates (Build)	Grant	Primary Output
Wave 1	Foundation	Mar 30 – Apr 6	$3,000	Club NFT + Match Engine on testnet
Wave 2	Financial Layer	Apr 8 – May 8*	$5,000	Sealed betting + encrypted fan tokens
Wave 3	Core Protocol	Apr 8 – May 8*	$12,000	Encrypted vAMM perps + full integration
Wave 4	Frontend + UX	May 11 – May 20	$14,000	Complete dApp, all encrypted UX flows
Wave 5	Launch Prep	May 23 – Jun 1	$16,000	Audit, mainnet prep, ecosystem showcase

** Wave 2 and Wave 3 run concurrently during the marathon period (Apr 8 – May 8). Wave 3 deliverables depend on Wave 2 contracts being complete.*

Wave 1 — Foundation

March 30 – April 6, 2025 (Build) | April 6–8 (Evaluation) | **Grant allocation: $3,000**

Objective

Deploy the game layer: Club NFTs, the Match Engine oracle, the $DRIP utility token, and the Upgrade system. All contracts live on Arbitrum Sepolia (Fhenix CoFHE). The FHE environment is configured and FHE-native formation submission is working from day one.

Deliverables

#	Deliverable	Success Metric	Evaluation Signal
1	ClubNFT.sol deployed and verified	Arbiscan-verified contract; mint tx produces valid randomised stats (10–100)	Contract address + sample mint tx hash
2	MatchEngine.sol deployed and wired	Challenge/accept flow executes; outcome calls ClubNFT.recordMatch	Match event on testnet; W/L record updated
3	Encrypted formation submission	euint8 formation submitted encrypted; decrypted at match resolution	FHE decrypt event in match resolution tx
4	DripToken.sol + UpgradeManager.sol	Win mints DRIP; upgrade burns DRIP at correct formula; fee to treasury	Token transfer events; cost formula verified
5	Integration test suite passing	npx hardhat test covers full Wave 1 flow with FHE mock	Test output screenshot / CI log
6	Hardhat FHE environment documented	.env.example, README setup instructions, @cofhe/hardhat confirmed working	Reviewer can clone and run tests in < 10 min

FHE Components in Wave 1

Component	FHE Operation	Privacy Guarantee
Formation selection	FHE.decrypt(euint8) at match resolution	Formation unknown to opponent until kick-off
Dev environment	FHE mock via @cofhe/hardhat	All FHE types testable locally without testnet

What Wave 1 is NOT expected to deliver

- Betting or perps (Wave 2/3)
- Frontend (Wave 4)
- Mainnet deployment (Wave 5)

Wave 2 — Financial Layer

April 8 – May 8, 2025 (Build) | April 6–8 (Evaluation) | **Grant allocation: $5,000**

Objective

Deploy the sealed-bid betting pool and the fan token bonding curve — both with FHE encryption on all financial inputs. These are the first user-facing demonstrations that encrypted compute changes the product fundamentally.

Deliverables

#	Deliverable	Success Metric	Evaluation Signal
1	BettingPool.sol (sealed-bid)	Encrypted bets accepted; pool tally not readable during live betting; correct payout at settlement	3-wallet test: place encrypted bets, settle, claim — payout distribution correct
2	Commit-then-prove collateral model	Encrypted amount cannot exceed plaintext collateral; excess forfeited	Attack test: submit euint32 larger than collateral — tx fails or excess forfeited
3	FanTokenFactory + BondingCurve	Club with 10 wins launches fan token; buy/sell with encrypted trade size works	Fan token deployed; buy tx with encrypted amount; royalty arrives at club owner
4	Slippage protection on encrypted trades	Revert when actual price impact exceeds user's maxSlippageBps	Test: set tight slippage, submit large encrypted buy — tx reverts
5	Integration: MatchEngine settles BettingPool	Match resolution triggers pool settlement automatically	Settlement event in match resolution tx
6	Wave 2 integration test suite	Full flow: bet → match → settle → claim, with FHE mock	Test output / CI log

FHE Components in Wave 2

Component	FHE Operation	Privacy Guarantee
Bet amount	FHE.add(euint64) for running tally	No one reads pool distribution before settlement
Bet side	ebool stored encrypted per user	Bet direction sealed until match resolves
Fan token trade size	euint32 encrypted at submission	Whale buys cannot be sandwich-attacked
Collateral check	FHE comparison at claim: encrypted bet <= plaintext collateral	Prevents encrypted-amount inflation attack

Wave 3 (Marathon) — Core Protocol — Encrypted Perpetuals

April 8 – May 8, 2025 (Build) | May 8–11 (Evaluation) | **Grant allocation: $12,000**

Objective

Deploy the encrypted perpetuals vAMM — the most technically sophisticated FHE component in the protocol. Positions are stored as ciphertexts. Funding rate is computed on encrypted aggregate OI. Liquidation is triggered by on-chain FHE comparison, not plaintext visibility. Wire all contracts together through MatchEngine as the settlement hub.

Deliverables

#	Deliverable	Success Metric	Evaluation Signal
1	PerpVault.sol — encrypted positions	euint64 size, euint8 leverage, euint64 liqPrice, ebool isLong stored per position	Position struct readable only via user permit; no plaintext position data in events
2	Encrypted open/close flow	openPosition and closePosition execute with FHE-encrypted inputs	Open + close tx on testnet; PnL computed correctly against collateral
3	FHE liquidation trigger	FHE.lt(equity, threshold) returns ebool; contract executes liquidation only on true result	Undercollateralised position liquidated; liquidator earns 1% without position size being emitted
4	Encrypted aggregate OI + funding rate	FHE.add over all position sizes per side; rate applied correctly	Funding rate event with correct sign; aggregate OI not exposed in plaintext
5	Match-triggered index settlement	PerpVault.settleIndex called by MatchEngine; win/draw/loss moves index to 75/50/25	Index update event after match resolution
6	Keeper scripts (off-chain)	fundingRate.ts and liquidationBot.ts execute on testnet	Keeper README; sample keeper run log
7	Full protocol integration	Match resolution triggers ClubNFT + BettingPool + PerpVault + DripToken in sequence	Single match resolution tx triggers all settlement events; gas < 500k
8	Wave 3 integration test suite	Perp open/close/liquidate/settle — all paths covered	Test output / CI log; coverage > 90%

FHE Components in Wave 3

Component	FHE Operation	Privacy Guarantee
Position size	euint64 stored and used in PnL calculation	Adversaries cannot read position size to hunt liquidations

Component	FHE Operation	Privacy Guarantee
Leverage	euint8 stored; used in liqPrice FHE arithmetic	Leverage not visible on-chain
Liquidation price	euint64 computed via FHE arithmetic at open	liqPrice not emittable; cannot be front-run
Direction (long/short)	ebool stored per position	Direction sealed — no directional bias signal
Aggregate OI	FHE.add() over all position sizes	Total OI computed without revealing individual positions
Liquidation check	FHE.lt(equity, threshold) → ebool only	Liquidator earns bounty; never sees position details

Why Wave 3 earns the highest internal grant allocation

The encrypted perp system is the primary technical differentiator. Transparent perp markets on EVM have failed to attract serious capital because position hunting and OI-based front-running are trivial. FHE.lt() returning only a boolean — proving undercollateralisation without revealing the position — is a novel primitive that does not exist in any live EVM protocol today.

Wave 4 — Frontend + UX

May 11 – May 20, 2025 (Build) | May 20–23 (Evaluation) | **Grant allocation: $14,000**

Objective

Ship the complete dApp with all encrypted UX flows working end-to-end on Arbitrum Sepolia (Fhenix CoFHE). Users should be able to complete every protocol action — mint, match, bet, trade fan tokens, open and close a perp position — without understanding FHE. The privacy is architectural, not a UX burden.

Deliverables

#	Deliverable	Success Metric	Evaluation Signal
1	Next.js dApp scaffold + wallet connection	MetaMask connects to Arbitrum Sepolia; @cofhe/react hooks available	Wallet connect demo on testnet
2	The Graph subgraph indexing all events	Match history, club stats, bonding curve price points, funding rate snapshots queryable via GraphQL	GraphQL playground link; sample queries documented
3	Club dashboard (/clubs)	All clubs visible with stats, W/L, upgrade UI, challenge button	Live page on testnet with real data
4	Match room (/match/[id])	Formation submission (encrypted), bet placement (encrypted), result reveal, claim button	End-to-end match from two wallets: formation + bet + settle + claim
5	Fan token trading (/tokens/[clubId])	Buy/sell with encrypted amounts; royalty tracker for club owners	Live buy tx with encrypted amount; royalty balance updates
6	Perpetuals page (/perps)	Open/close with encrypted inputs; own positions visible via decrypt permit; funding rate display	Open perp tx; position visible to owner only; hidden from second wallet
7	Encrypted UX copy standard	Consistent messaging across all FHE interactions; permit flow explained to user	Copy library in codebase; reviewer confirms UX is clear on all encrypted actions
8	Mobile responsive + QA	All pages usable on 375px; no broken states; optimistic updates on writes	Mobile screenshot audit; error state walkthrough

Key UX principle for evaluation

The encrypted flows must not feel broken or slow to a user unfamiliar with FHE. Specifically: placing an encrypted bet should feel like placing any on-chain bet — the privacy is communicated as a feature ('Your bet is sealed. Running odds are intentionally hidden.'), not an error or a limitation.

Wave 5 — Audit, Mainnet Prep, Showcase

May 23 – June 1, 2025 (Build) | June 1–5 (Evaluation) | **Grant allocation: $14,000 + $2,000 bonus**

Objective

Prepare Drip Protocol for mainnet deployment and ecosystem showcase. Security review completed, deployment infrastructure production-grade, documentation complete, demo video ready for Fhenix community and ecosystem partners.

Deliverables

#	Deliverable	Success Metric	Evaluation Signal
1	Internal security audit completed	All Slither high/medium findings addressed; reentrancy guards verified; FHE overflow analysis done	Slither output + remediation notes in AUDIT.md
2	Test coverage > 90%	Hardhat coverage report across all contracts	Coverage report screenshot / HTML output
3	Mainnet deployment scripts	Multi-sig deployer; 48h timelock on ProtocolConfig; deploy-mainnet.ts idempotent	Deploy script dry-run log; Gnosis Safe address
4	Complete documentation	architecture.md, fhe-design.md, user-guide.md, keeper-guide.md	docs/ directory in repo; reviewer can run keeper from guide
5	Demo video (< 5 min)	Full user journey on testnet: mint → match (with encrypted formation) → encrypted bet → settle → fan token buy → open perp → close perp	YouTube/Loom link submitted to Fhenix buildathon
6	AKINDO submission complete	Project page live on AKINDO with repo, demo, description	AKINDO project URL
7	Ecosystem showcase ready	Slides and live demo prepared for NY Tech Week / Fhenix community event	Slide deck link; live demo working on day of showcase

Bonus criteria ($2,000 bonus allocation)

- Protocol deployed on Fhenix mainnet (not just Arbitrum Sepolia testnet)
- External security review completed (independent auditor, written report)
- Partnership or integration announcement with another Fhenix ecosystem project

Evaluation Criteria — All Waves

The Fhenix buildathon evaluates on creative and effective application of privacy-first architecture. For Drip Protocol, the evaluation bar at each wave is:

Technical (50%)

- FHE types used correctly — euint32/euint64/ebool/euint8 for the right data, not as boilerplate
- Access control model sound — decryption only where appropriate (user permit or protocol settlement)
- No plaintext leakage — encrypted values not emitted in events, not returned in public getters
- Integration tight — MatchEngine triggers all downstream settlement atomically
- Gas within target — match resolution < 500k gas

Product (30%)

- Privacy is the product feature, not a technical footnote — UX communicates it as value
- Full user journey completable — from mint to perp settlement without touching a wallet extension manually beyond signing
- The game layer and financial layer are genuinely integrated — the match is not decorative

Documentation + Reproducibility (20%)

- Reviewer can clone and run tests in < 10 minutes
- FHE design choices documented — why each field is encrypted, what the privacy guarantee is
- Deploy scripts idempotent and annotated

Risk Register

Risk	Severity	Mitigation
FHE gas costs spike, making match resolution unaffordable	Medium	Profile early; pre-compute liqPrice off-chain if needed; batch settlement
@cofhe/react hooks incompatible with Next.js 14 App Router	Low	Test scaffold in Issue #15 first; fallback to Pages Router if needed
Encrypted amount inflation attack on BettingPool	High	Commit-then-prove pattern (Issue #7); attack test in acceptance criteria

Risk	Severity	Mitigation
FHE decrypt latency in frontend — UX feels broken	Medium	Optimistic UI; loading states; communicate FHE decrypt as a feature not a bug
Fhenix Arbitrum Sepolia testnet instability	Low	Local FHE mock for all development; testnet only for final integration tests
Wave 3 FHE arithmetic complexity overruns timeline	Medium	Perps are isolated contract; can demo with partial FHE (size encrypted, leverage plaintext) if needed

Drip Protocol milestones are tracked publicly on GitHub. Wave evaluation submissions link directly to commit hashes and testnet transaction receipts.